

SURENDRANATH COLLEGE

Mode of Examination: Online

M.Sc. (Computer Science)

Semester-II Intermediate Examination, 2020

Subject: Computer Science

Paper: CSM202

(Design and Analysis Of Algorithms)

Full Marks: 50

Time: 2:30 Hours

Date: 09.12.2020

Duration: 11:00 AM – 1:30PM

Please follow the following instructions carefully.

Students have to attempt both sections A and B .

The figures in the margin indicate full marks.

Candidates are required to answer in their own words as far as applicable.

Each Page of answer scripts should have your examination Roll Number on the right side top corner of your answer script and the page number.

The name of the scanned copy of the answer script will be of the following format:

MSC_CSM202_Roll-Number.pdf

(Example: MSC_CSM202_115/CSC/181001)

The subject of the mail should be answer-script file name.

The scanned answer script is to be sent to taniya.snc.cuexam@gmail.com

No answer script is valid if received after **1:30 PM on the Date of Examination.**

Write the answers with black/blue ink ball pen.

Section-A3. Answer **any five** questions:

2X5

- (a) Give an example of quadratic probing for a hash table of size m , which generates a permutation of 0 to $m-1$.
- (b) Define Universal Hashing.
- (c) A chained hash table has an array size of 100. What is the maximum number of entries that can be placed in the table ?
- (d) What is the significance of load factor of a Hash Table?
- (e) "If X is NP-complete, then X is solvable in polynomial time if and only if $P = NP$ " True or False ? Justify.
- (f) Define Clique problem.
- (g) Define the complexity NP-complete.
- (h) Define Travelling Salesman Problem.

4. Answer **any two** questions:

5X2

- (a) What is the expected number of probes for unsuccessful search in an open addressed hash table with load factor α ? (5)
- (b) (i) Define quadratic probing .
(ii) Show that this quadratic probing indeed probes all slots. (2+3)
- (c) (i) State Cook's theorem.
(ii) Given that 3-CNF problem is NP Complete, prove that Clique problem is NP Complete. (2+3)
- (d) (i) Define perfect hashing.
(ii) Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first 3 insertions? (2+3)
- (e) (i) What is the aim of Backtracking?

(3)

- (ii) Consider a 4X4 chessboard like structure where you have to place 4 queens such that no one attacks each other. Give an algorithm to find a solution.

(2+3)

Section-B

1. Answer **any five** questions:

2X5

- (a) Let $f(n)$ and $g(n)$ be positive functions, Prove that if $f(n)$ is $O(g(n))$ then $g(n)$ is $\Omega(f(n))$.
- (b) Define Amortized Cost.
- (c) State max flow- min cut theorem.
- (d) Consider an alternative to binary search called ternary search that divides the input array into three equal parts and recursively searches one of these three segments. What can we say about the asymptotic worst case complexity of ternary search?
- (e) Define Longest Common Subsequence problem.
- (f) Write the difference between the Greedy method and Dynamic programming.
- (g) What is best running time to sort n integers in the range 0 to $n^2 - 1$?
- (h) What is augmentation and augmentation path?

2. Answer **any four** questions:

5X4

- (a) (i) An image is represented as an $N \times N$ array A of pixels. There is a function that transforms the image as follows:
- The function scans each pixel $A[i][j]$. Depending on the current value of $A[i][j]$, some values in row i and column j are updated. It is possible that all values in row i and column j are updated. It is also possible that no values are updated.
 - Updating a pixel takes time $O(\log N)$.
 - Each pixel in the image is updated at most once by the function.

What is the best upper bound you can estimate for the total time taken across all the updates made by the function?

(4)

(ii) Solve the following recurrence .

$$T(n) = 4T(n/2) + n^2 \text{ where } n > 1 \text{ and } T(1) = 1 \quad (3+2)$$

(b) Write the algorithm for merge sort. Given $\log n$ sorted list each of size $n/\log n$. What is the total time required to merge them into one single list. (3+2)

(c) Write algorithms for operations, MAKE-SET, FIND-SET and UNION on disjoint sets represented as rooted trees such that cost of m MAKE-SET, FIND-SET and UNION operations, n of which are MAKE-SET operations is, $O(m \log n)$. (5)

(d) (i) What is the minimum number of comparison required to find maximum and minimum among n integers?(Your expression should hold for both n even and odd)

(ii) "Finding the least element is asymptotically faster than finding the k th ($k > 1$) element from an unsorted array" True or False ? Justify (3+2)

(e) You are given set of eleven half open interval $[I_a, I_b)$ as follows, Find

Interval	1	2	3	4	5	6	7	8	9	10	11
I_a	1	3	0	5	3	5	6	8	8	2	12
I_b	4	5	6	7	9	9	10	11	12	14	16

a maximum set of non overlapping intervals (intervals $[a,b)$ and $[c,d)$ are non overlapping if either $c \geq b$ or $a \geq d$) (5)

(f) Explain Prim's algorithm and its time complexity. Analyze it with an example.(5)

(g) In a non-negatively edge-weighted undirected graph $G=(V,E)$, an $(s-t)$ cut is a vertex partition $(S, V \setminus S)$ with $s \in S$ and $t \in V \setminus S$. Weight of the cut is the sum of weights of the edges $\{(x,y) | x \in S, y \in V \setminus S\}$. Define a suitable flow network so that the maximum flow computation in it can give the weight of minimum weight $(s-t)$ cut in G . Justify. (5)
